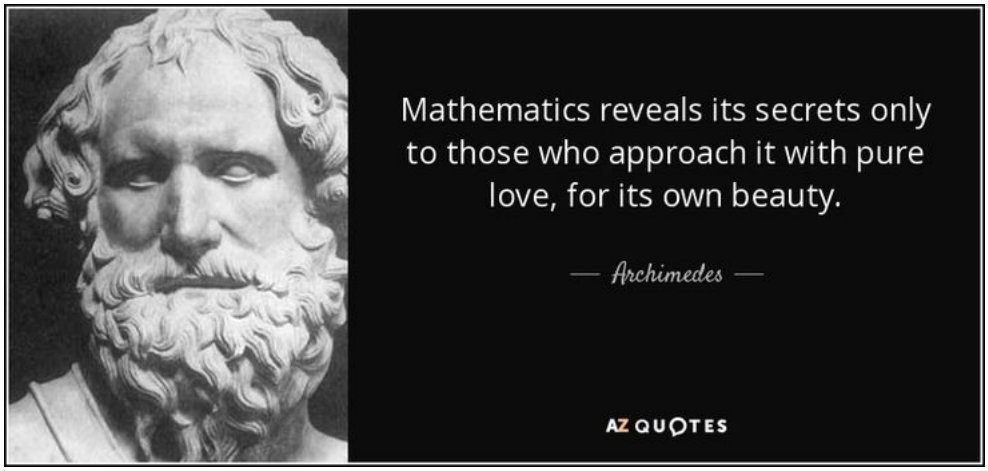
Chapter 3
Sequences of Real Numbers

Note 2

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3.2 : LIMIT THEOREMS

One of the biggest challenges we face (both for theory and applications) is deciding whether or not a given sequence converges. Once we know that it converges, we can often use other techniques to approximate or evaluate its limit.

One way to identify convergent sequences is by comparing a sequence whose convergence is in doubt with another whose convergence property is already known (see **Example 1**). The following result is the first of many theorems that addresses this issue.

Theorem 3.2.1 : Squeeze Theorem

Suppose that $\{x_n\}, \{y_n\}$, and $\{w_n\}$ are real sequences.

(i). If $x_n \to a$ and $y_n \to a$ (the SAME a) as $n \to \infty$, and if there is an $N_0 \in \mathbb{Z}^+$ such that

$$x_n \leq w_n \leq y_n \quad \text{for } n \geq N_0,$$

then $w_n \to a$ as $n \to \infty$.

(ii). If $x_n \to 0$ as $n \to \infty$ and $\{y_n\}$ is bounded, then $x_n y_n \to 0$ as $n \to \infty$.

Proof.

(i).

- Let $\varepsilon > 0$. Since $\{x_n\}$ and $\{y_n\}$ converge to a, use **Definition 3.1.1** and **Theorem 3.1** to choose $N_1, N_2 \in \mathbb{Z}^+$ such that $n \geq N_1$ implies $-\varepsilon < x_n - a < \varepsilon$ and $n \geq N_2$ implies $-\varepsilon < y_n - a < \varepsilon$.
- Set $N = \max\{N_0, N_1, N_2\}$. If $n \geq N$, we have by the hypothesis and the choice of N_1 and N_2 that

$$a - \varepsilon < x_n \leq w_n \leq y_n < a + \varepsilon \ ;$$

that is, $|w_n - a| < \varepsilon$ for $n \geq N$. We conclude that $w_n \to a$ as $n \to \infty$.

(ii).

- Suppose that $x_n \to 0$ and that there is an $M > 0$ such that $|y_n| \leq M$ for $n \in \mathbb{Z}^+$.
- Let $\varepsilon > 0$ and choose an $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $|x_n| < \dfrac{\epsilon}{M}$. Then $n \geq N$ implies

$$|x_n y_n| < M \cdot \frac{\epsilon}{M} = \varepsilon \ .$$

We conclude that $x_n y_n \to 0$ as $n \to \infty$.

□

The following example shows how to use the Squeeze Theorem to find the limit of a complicated sequence by ignoring its "less important" factors.

Example 1 Find $\displaystyle\lim_{n\to\infty} \frac{1}{2^n} \cos\left(n^3 - n^2 + n - 13\right)$

Solution : Since $|\cos x| \leq 1$ for all $x \in \mathbb{R}$, we have

$$-\frac{1}{2^n} \leq \frac{1}{2^n} \cos\left(n^3 - n^2 + n - 13\right) \leq \frac{1}{2^n} \ .$$

Since $\dfrac{1}{2^n} \to 0$ as $n \to \infty$, by Squeeze theorem we have

$$\lim_{n\to\infty} \frac{1}{2^n} \cos\left(n^3 - n^2 + n - 13\right) = 0 \ .$$

The Squeeze Theorem can also be used to construct convergent sequences with certain properties. To illustrate how this works, we now establish a result that connects suprema and infima with convergent sequences.

Theorem 3.2.2 : Let $E \subset \mathbb{R}$. If E has a finite supremum (respectively, a finite infimum), then there is a sequence $\{x_n\} \in E$ such that $x_n \to \sup E$ (respectively, a sequence $\{y_n\} \in E$ such that $y_n \to \inf E$) as $n \to \infty$.

Proof.

- Suppose that E has a finite supremum.
- For each $n \in \mathbb{Z}^+$, by the **Approximation Property for Suprema**, choose a sequence $\{x_n\} \in E$ such that
$$\sup E - \frac{1}{n} < x_n \leq \sup E \ .$$
- Then by the **Squeeze Theorem**, it clearly follows that $x_n \to \sup E$ as $n \to \infty$.
- Similarly, there is a sequence $\{y_n\} \in E$ such that $y_n \to \inf E$.

□

Here is another result that helps to evaluate limits of specific sequences. This one works by viewing complicated sequences in terms of simpler components.

Theorem 3.2.3 : Suppose that $\{x_n\}$ and $\{y_n\}$ are real sequences and that $\alpha \in \mathbb{R}$. If $\{x_n\}$ and $\{y_n\}$ are convergent, then

(i). $\lim\limits_{n\to\infty} (x_n + y_n) = \lim\limits_{n\to\infty} x_n + \lim\limits_{n\to\infty} y_n$.

(ii). $\lim\limits_{n\to\infty} (\alpha x_n) = \alpha \lim\limits_{n\to\infty} x_n$.

(iii). $\lim\limits_{n\to\infty} (x_n y_n) = \left(\lim\limits_{n\to\infty} x_n\right)\left(\lim\limits_{n\to\infty} y_n\right)$

(iv). If $y_n \neq 0$ and $\lim_{n\to\infty} y_n \neq 0$, then
$$\lim_{n\to\infty} \frac{x_n}{y_n} = \frac{\lim_{n\to\infty} x_n}{\lim_{n\to\infty} y_n} \ .$$
(In particular, all these limits exist.)

Proof. Suppose that $x_n \to x$ and $y_n \to y$ as $n \to \infty$.

(i).

- Let $\varepsilon > 0$ and choose $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $|x_n - x| < \dfrac{\epsilon}{2}$ and $|y_n - y| < \dfrac{\epsilon}{2}$.
- By the **triangle inequality**, it follows that

$$\begin{aligned} |(x_n + y_n) - (x + y)| &\leq |x_n - x| + |y_n - y| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

(ii). It suffices to show that $\alpha x_n - \alpha x \to 0$ as $n \to \infty$. But $x_n - x \to 0$ as $n \to \infty$, hence by the **Squeeze Theorem**, $\alpha(x_n - x) \to 0$ as $n \to \infty$.

Parts **(iii)** and **(iv)** are left as Exercises. □

Theorem 3.2.3 can be used to evaluate limits of sums, products, and quotients. Here is a typical example.

Example 2 Find $\lim\limits_{n\to\infty} \dfrac{(n^3 + n^2 - 1)}{(1 - 3n^3)}$.

Solution. Multiplying the numerator and denominator by $1/n^3$, we find that

$$\frac{n^3 + n^2 - 1}{1 - 3n^3} = \frac{1 + (1/n) - (1/n^3)}{(1/n^3) - 3} \ .$$

As $n \to \infty$, by **Theorem 4.2.3** (i), (ii), and (iv), it follows that

$$\lim_{n\to\infty} \frac{n^3 + n^2 - 1}{1 - 3n^3} = \frac{1 + 0 - 0}{0 - 3} = -\frac{1}{3} \ .$$

Note : The sequence $\{\log n\}_{n\in\mathbb{Z}^+}$ fails to converge in a different way than $\{n(-1)^n\}_{n\in\mathbb{Z}^+}$ does. Indeed, the terms $\log n$ get steadily larger as $n \to \infty$, but the terms $n(-1)^n$ bounce back and forth between large positive values and large negative values. It is sometimes convenient to emphasize this difference by generalizing limits to include extended real numbers.

Definition 3.2.1

Let $\{x_n\}$ be a sequence of real numbers.

(i). $\{x_n\}$ is said to **diverge** to $+\infty$ (notation: $x_n \to +\infty$ as $n \to \infty$ or $\lim_{n\to\infty} x_n = +\infty$ if and only if for each $M \in \mathbb{R}$ there is an $N \in \mathbb{Z}^+$ such that

$$n \geq N \quad \text{implies} \quad x_n > M.$$

(ii). $\{x_n\}$ is said to **diverge** to $-\infty$ (notation: $x_n \to -\infty$ as $n \to \infty$ or $\lim_{n\to\infty} x_n = -\infty$ if and only if for each $M \in \mathbb{R}$ there is an $N \in \mathbb{Z}^+$ such that

$$n \geq N \quad \text{implies} \quad x_n < M.$$

Note : By the above definition $x_n \to +\infty$ if and only if given $M \in \mathbb{R}$, $x_n > M$ for sufficiently large n; that is, eventually x_n exceeds every number M (no matter how large and positive M is). Similarly, $x_n \to -\infty$ if and only if x_n eventually is less than every number M (no matter how large and negative M is).

Theorem 3.2.4 : Suppose that $\{x_n\}$ and $\{y_n\}$ are real sequences such that $x_n \to +\infty$ (respectively, $x_n \to -\infty$) as $n \to \infty$.

(i). If y_n is bounded below (respectively, y_n is bounded above), then

$$\lim_{n\to\infty} (x_n + y_n) = +\infty \quad \left(\text{respectively, } \lim_{n\to\infty} (x_n + y_n) = -\infty\right) .$$

(ii). If $\alpha > 0$, then

$$\lim_{n\to\infty} (\alpha x_n) = +\infty \quad \left(\text{respectively, } \lim_{n\to\infty} (\alpha x_n) = -\infty\right) .$$

(iii). If $y_n > M_0$ for some $M_0 > 0$ and all $n \in \mathbb{Z}^+$, then

$$\lim_{n\to\infty} (x_n y_n) = +\infty \quad \left(\text{respectively, } \lim_{n\to\infty} (x_n y_n) = -\infty\right) .$$

(iv). If $\{y_n\}$ is bounded and $x_n \neq 0$, then

$$\lim_{n\to\infty} \frac{y_n}{x_n} = 0 .$$

Proof. We suppose for simplicity that $x_n \to +\infty$ as $n \to \infty$.

(i).
- By hypothesis, $y_n \geq M_0$ for some $M_0 \in \mathbb{R}$.
- Let $M \in \mathbb{R}$ and set $M_1 = M - M_0$.
- Since $x_n \to +\infty$, choose $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $x_n > M_1$.
- Then $n \geq N$ implies $x_n + y_n > M_1 + M_0 = M$.

(ii).
- Let $M \in \mathbb{R}$ and set $M_1 = \dfrac{M}{\alpha}$.
- Choose $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $x_n > M_1$.
- Since $\alpha > 0$, we conclude that $\alpha x_n > \alpha M_1 = M$ for all $n \geq N$.

(iii).
- Let $M \in \mathbb{R}$ and set $M_1 = \dfrac{M}{M_0}$.
- Choose $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $x_n > M_1$.
- Then $n \geq N$ implies $x_n y_n > M_1 M_0 = M$.

(iv).
- Let $\varepsilon > 0$. Choose $M_0 > 0$ such that $|y_n| \leq M_0$ and $M_1 > 0$ so large that $\dfrac{M_0}{M1} < \varepsilon$.
- Choose $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $x_n > M_1$. Then $n \geq N$ implies

$$\left|\frac{y_n}{x_n}\right| = \frac{|y_n|}{x_n} < \frac{M_0}{M1} < \varepsilon \ .$$

□

If we adopt the conventions

$$x + \infty = \infty, \quad x - \infty = -\infty, \quad x \in \mathbb{R}$$
$$x \cdot \infty = \infty, \quad x \cdot (-\infty) = -\infty, \quad x > 0$$
$$x \cdot \infty = -\infty, \quad x \cdot (-\infty) = \infty, \quad x < 0$$
$$\infty + \infty = \infty, \quad -\infty - \infty = -\infty$$
$$\infty \cdot \infty = (-\infty) \cdot (-\infty) = \infty, \quad \text{and} \quad \infty \cdot (-\infty) = (-\infty) \cdot \infty = -\infty$$

then **Theorem 3.2.4** contains the following corollary.

Corollary : Let $\{x_n\}, \{y_n\}$ be real sequences and α, x, y be extended real numbers. If $x_n \to x$ and $y_n \to y$, as $n \to \infty$, then

$$\lim_{n\to\infty} (x_n + y_n) = x + y$$

provided that the right side is not of the form $\infty - \infty$, and

$$\lim_{n\to\infty} (\alpha x_n) = \alpha x, \quad \lim_{n\to\infty} (x_n y_n) = xy$$

provided that none of these products is of the form $0 \cdot \pm\infty$.

Theorems 3.2.3 and **3.2.4** show how the limit sign interacts with the algebraic structure of $\mathbb{R}$ (Namely, the limit of a sum (product, quotient) is the sum (product, quotient) of the limits). The following theorem shows how the limit sign interacts with the order structure of $\mathbb{R}$.

Theorem 3.2.5 : (Comparison Theorem)

Suppose that $\{x_n\}$ and $\{y_n\}$ are convergent sequences. If there is an $N_0 \in \mathbb{Z}^+$ such that

$$x_n \leq y_n \quad \text{for } n \geq N_0 \tag{1}$$

then

$$\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n \ . \tag{2}$$

In particular, if $x_n \in [a, b]$ converges to some point c, then c must belong to $[a, b]$

Proof.

- Suppose that the first statement is false; that is, that (1) holds but $x := \lim_{n\to\infty} x_n > y := \lim_{n\to\infty} y_n$.
- Set $\varepsilon = \dfrac{(x-y)}{2}$.
- Choose $N_1 > N_0$ such that $|x_n - x| < \varepsilon$ and $|y_n - y| < \varepsilon$ for $n \geq N_1$. Then for such an n,

$$x_n > x - \varepsilon = x - \left(\frac{x-y}{2}\right) = y + \left(\frac{x-y}{2}\right) = y + \varepsilon > y_n$$

  which contradicts (1). This proves the first statement.
- We conclude by noting that the second statement follows from the first, since $a \leq x_n \leq b$ implies $a \leq c \leq b$.

□

Remarks :

- One way to remember **Theorem 3.2.5** is that it says the limit of an inequality is the inequality of the limits, provided these limits exist.

- We shall call this process "taking the limit of an inequality".

- Since $x_n < y_n$ implies $x_n \leq y_n$, **Theorem 3.2.5** contains the following corollary:

  Corollary : If $\{x_n\}$ and $\{y_n\}$ are convergent sequences, then

  $$x_n < y_n \ , \quad n \geq N_0 \ , \quad \text{imply} \quad \lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n \ .$$

- It is important to notice that this result is false if $\leq$ is replaced by $<$; that is,

  $$x_n < y_n \ , \quad n \geq N_0 \ , \quad \text{does NOT imply that} \quad \lim_{n\to\infty} x_n < \lim_{n\to\infty} y_n \ .$$

  For example, $\dfrac{1}{n^2} < \dfrac{1}{n}$, but the limits of these sequences are equal.

EXERCISES

(1). Determine which of the following statements are true and which are false. Prove the true ones and provide counterexamples for the false ones.

(a). If $x_n \to \infty$ and $y_n \to -\infty$, then $x_n + y_n \to 0$ as $n \to \infty$

(b). If $x_n \to -\infty$, then $\dfrac{1}{x_n} \to 0$ as $n \to \infty$

(c). If $x_n \to 0$, then $\dfrac{1}{x_n} \to \infty$ as $n \to \infty$

(d). If $x_n \to \infty$, then $\left(\dfrac{1}{2}\right)^{x_n} \to 0$ as $n \to \infty$

(2). Prove that each of the following sequences converges to zero.

(a). $x_n = \dfrac{1}{n}\sin(\log n + n^5 + e^{n^2})$

(b). $x_n = \dfrac{2n}{(n^2+\pi)}$

(c). $x_n = \dfrac{(\sqrt{2n}+1)}{(n+\sqrt{2})}$

(d). $x_n = \dfrac{n}{2^n}$

(3). Use the Definition in page 4 to prove that each of the following sequences diverges to $+\infty$ or to $-\infty$.

(a). $x_n = n^2 - n$

(b). $x_n = n - 3n^2$

(c). $x_n = \dfrac{n^2+1}{n}$

(d). $x_n = n^2\,(2 + \sin(n^3 + n + 1))$

(4). Find the limit (if it exists) of each of the following sequences.

(a). $x_n = \dfrac{(2+3n-4n^2)}{(1-2n+3n^2)}$

(b). $x_n = \dfrac{(n^3+n-2)}{(2n^3+n-2)}$

(c). $x_n = \sqrt{3n+2} - \sqrt{n}$

(d). $x_n = \dfrac{(\sqrt{4n+1}-\sqrt{n-1})}{(\sqrt{9n+1}-\sqrt{n+2})}$